Lucas de Linhares

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Education

Polytechnic University of Viana do Castelo — Viana do Castelo, PT
Bachelor's Degree in Network and Computer Systems Engineering — *Sep. 2025 - Present*

Polytechnic University of Viana do Castelo — Viana do Castelo, PT
Higher Professional Technical Course in Information Systems Technologies and Programming — *Sep. 2023 - May 2025*

Monserrate Secondary School — Viana do Castelo, PT
Professional Course in Electronics, Automation and Control — *Sep. 2020 - Jul. 2023*

Experience

Researcher — Jun. 2026 - Present
Polytechnic University of Viana do Castelo — *Viana do Castelo, PT*

- Conducting a systematic literature review on Zero Trust Architecture (ZTA) in IT/OT environments within Industry 5.0
- Developed an AI-assisted platform that automates keyword extraction, thematic clustering, synthesis matrix generation, and research gap identification using LLMs
- Designed a parallel processing pipeline with intelligent caching to accelerate literature analysis and reduce redundant LLM processing

Software Engineer Intern — Feb. 2025 - Jun. 2025
JCC - Diagnostic Imaging — *Viana do Castelo, PT*

- Developed a company-wide document management and workflow platform using Rust, Actix-Web, Svelte and MySQL
- Designed a fully configurable architecture for users, permissions, document types and workflows
- Implemented RBAC, notifications, due-date tracking, vacation management and internal communication features

Machine Learning Intern — May 2023 - Jul. 2023
Ctrl + Alt DEV — *Viana do Castelo, PT*

- Developed predictive maintenance models using TensorFlow, Scikit-Learn, Pandas and NumPy for industrial machinery

Software Engineer Intern — May 2022 - Jul. 2022
Ctrl + Alt DEV — *Viana do Castelo, PT*

- Enhanced a machine management platform using Node.js, AdonisJS, React and PostgreSQL
- Integrated PLCs with the existing platform and contributed to feature development and debugging

Projects

Photography E-commerce Platform | *Rust, Axum, SvelteKit, TypeScript, PostgreSQL, Stripe, S3* — Nov. 2020 - Present

- Designed, built and maintain a production e-commerce platform handling thousands of monthly transactions
- Developed an optimized media pipeline with AVIF conversion, dynamic watermarking and scalable S3-compatible storage
- Built a complete administration platform with analytics, sales reporting and content management
- Achieved perfect Lighthouse scores through frontend and backend optimization

Macro Center | *Rust, Tauri, Svelte* — Mar. 2023 - Present

- Developed a cross-platform desktop automation platform with node-based visual programming
- Built a high-performance macro execution engine for keyboard and mouse automation
- Developed libmacrocenter, a Rust library with Python, Node.js, and Lua bindings for desktop automation, powering Macro Center.

Numera | *Rust, Image Processing, Computer Vision, Parallel Processing* — Jul. 2026 - Present

- Developed a high-performance desktop application for professional photographers, optimizing post-event image processing workflows
- Built a parallel image processing pipeline supporting configurable print-ready exports with resizing, DPI adjustment, JPEG quality, watermarks, overlays, borders, and image transformations
- Developing a computer vision pipeline to automatically recognize athlete and vehicle identification numbers from event photographs for automated image organization
- Implemented automated file organization tools that group and structure thousands of images based on recognized or assigned event numbers

Autism Support Automation Platform | *Go, PostgreSQL, Docker, React, Nginx* — Sep. 2023 - Jan. 2024

- Selected as 1 of 3 volunteers for an international accessibility project
- Diagnosed production issues, implemented fixes and documented deployment workflows

Technical Skills

Languages: Python, Rust, TypeScript, SQL, C, C++, Java, C#, Go
Frameworks: React, Next.js, Svelte, FastAPI, Django, Tauri, Axum, Angular, TensorFlow, Stripe
Developer Tools: Git, Docker, GitHub Actions, Netlify, Vercel, Linux, CMake, MySQL, Nginx, PostgreSQL
Spoken Languages: Portuguese (Native), English (Fluent), French (Professional Proficiency)